

Chapter 5 Working Capital Management

Let's Start

1

(a) It has to pay for extra space to store raw materials and finished goods.	✓
(b) It has to pay for extra labour costs involved in increased production.	✓
(c) It will collect trade receivables at the end of the fourth quarter.	✓

- 2 Kiddoland should finance its cash shortfall with a short-term bank loan. Since Kiddoland is expected to collect trade receivables at the end of the fourth quarter, the cash collected can be used to settle the short-term debt. If Kiddoland takes out a long-term loan instead, it has to pay unnecessary financing costs. This is because the interest rate on long-term loans is usually higher and the firm has to pay interest on idle funds.

Check Your Progress

- P5-1** (a) Accounts payable management
(b) Accounts receivable management
(c) Inventory management
(d) Cash management

- P5-2** (a) → (c) → (d) → (b)

- P5-3** (a) F The basic goal of cash management is maintaining adequate cash for business operations at a minimum cost.
(b) T
(c) T
(d) F Recipients prefer to shorten the collection float.

- P5-4** (a) Delaying cash outflows
(b) Matching the timing and amount of cash inflows and cash outflows
(c) Accelerating cash inflows (shortening the collection float)
(d) Accelerating cash inflows (encouraging payment in cash)

- P5-5** (a) F A cash budget is a budget for both cash receipts and payments.
(b) F A cash surplus is expected when the ending cash balance exceeds the minimum cash balance required.
(c) T
(d) T

- P5-6** (a) Character
(b) Capacity
(c) Conditions
(d) Capital
(e) Collateral
- P5-7** (a) T
(b) F A cash discount is usually expressed as a certain percentage of the invoice amount.
(c) T
(d) F The cash discount period is shorter than the credit period to encourage customers to pay early.
- P5-8** (a) Increase; Decrease; Decrease
(b) Decrease; Increase; Increase
(c) Increase; Increase; Decrease
(d) Increase; Decrease; Decrease
- P5-9** (a) F The cost of taking a cash discount is the loan interest or the investment income forgone.
(b) T
(c) T
(d) F A firm should take the cash discount if the cost savings resulting from the cash discount are greater than the loan interest cost.
- P5-10** (a) Yes; carrying cost
(b) No
(c) Yes; carrying cost
(d) Yes; ordering cost
- P5-11** (a) Total ordering cost of frozen food = $\$400 \times (2,000 \div 200) = \$4,000$
Total carrying cost of frozen food = $\$20 \times (200 \div 2) = \$2,000$
(b) Since the total ordering cost is greater than the total carrying cost, the supermarket should increase its ordering quantity of frozen food such that the total ordering cost will decrease and the total carrying cost will increase until a point at which they costs are equal.

P5-12 Economic order quantity of bottled water

$$= \sqrt{\frac{2(10,000)(170)}{20}}$$

= 413 bottles

Total inventory cost of bottled water

$$= \$170 \times (10,000 \div 413) + \$20 \times (413 \div 2)$$

$$= \$8,246$$

P5-13 (a) F The re-order level is the level of inventory at which a firm should place an order to replenish inventory.

(b) T

(c) F The greater the daily usage or sales of inventory, the higher the re-order level.

(d) T

P5-14 Re-order level

$$= (100 \times 5) + 60 = 560 \text{ units}$$

Let's Try

T5-1 We can lengthen the disbursement float in Hong Kong in the following ways:

- Mailing cheques just before weekends or long holidays, e.g., Christmas and Chinese New Year
- Issuing post-dated cheques
- Mailing cheques from remote districts such as outlying islands

(Any other reasonable answers)

T5-2 Firms sometimes choose not to accelerate cash inflows or delay cash outflows because the costs of doing so outweigh the benefits. For example:

- It is costly to set up collection centres in different major cities.
- In order to shorten the collection float, firms may have to employ couriers to collect cheques from customers.
- To discourage customers from paying by credit card, firms may charge fees for credit card payments. However, this may result in the loss of business.
- If a firm delays cash outflows too much, it may upset suppliers or creditors. They may refuse to grant good credit terms to the firm in future.

(Any other reasonable answers)

T5-3 To decide whether to grant trade credit to a customer, a firm should assess its creditworthiness in the short term. Therefore, the customer's liquidity ratios, which measure the customer's ability to repay its short-term liabilities when due, are the most useful for the firm. The firm can look at the customer's current ratio, quick ratio, inventory turnover, trade receivables turnover and trade payables turnover.

- T5-4** The firm may set up a meeting to negotiate debt extension agreements with debtors.
(Any other reasonable examples)

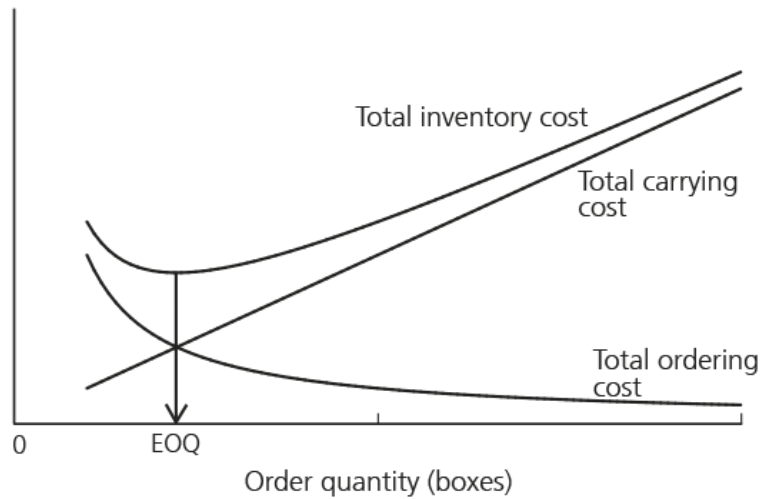
Answers to Assessment

Short Questions

- 5.1** (a) The basic goals of cash management include:
- Keeping adequate cash on hand for business operations 1 mark
 - Minimising the amount of idle cash so that the cost of holding cash is minimised 1 mark
- (b) The department store can encourage customers to pay with cash in the following ways:
- Issuing membership cards which allow customers to accumulate points by paying with cash or an Octopus card. The points can be redeemed for gifts.
 - Providing express checkout counters for cash purchases. Any two
 - Offering discounts for cash purchases. 2 marks
- (Any other reasonable examples)
- 5.2** (a) The online retailer purchases finished goods from suppliers on credit. Then, it sells the finished goods to consumers through the Internet. After collecting cash from sales, the online retailer pays its suppliers. 2 marks
- (b) The cash conversion cycle of the online retailer is relatively short because it sells its products online quickly and receives cash quickly after sales, meaning that inventory is converted into cash quickly. 2 marks
- 5.3** Factors that a bank should consider when setting its credit standards include:
- Capital: The amount of capital the business owner has invested in the business.
 - Capacity: The ability of the business to repay the debt.
 - Collateral: The collateral provided by the business as security in case of default.
 - Conditions: The overall economic conditions, the industry outlook of the business, as well as the intended purpose of the loan.
 - Character: The subjective judgment made by the lender regarding the creditworthiness of the business. Any three
2 marks each
- 5.4** Types of inventory cost under the Economic Order Quantity model include:
- Ordering costs: These are costs related to placing and receiving an order for inventory from suppliers. They include costs incurred from the clerical and administrative work involved transportation costs. 2 marks
 - Carrying costs: These are costs related to holding inventory for a certain time period (normally a year). They include storage expenses, insurance costs, the loss due to inventory deterioration and obsolescence, and the opportunity cost of the money tied up in inventory. 2 marks

- 5.5** Possible reasons for the supermarket chain to not take the cash discount:
- The firm does not have sufficient cash to settle payment earlier and the borrowing rate is higher than the annual rate on the cash discount. Hence, it is not beneficial for the firm to borrow money to settle payment earlier.
 - Business conditions are worsening and the firm expects that it will take longer to collect accounts receivable. As a result, the firm gives up the cash discount in order to maintain higher liquidity.
 - The firm needs extra cash in the near future to invest. As a result, the firm gives up its cash discount and holds the capital obtained through financing (i.e., the trade credit) for a longer period.
- Any two
2 marks each
- 5.6** (a) The liquidity of the swimwear manufacturer will increase because customers are more incentivised to repay earlier with a higher cash discount rate.
- 2 marks
- (b) The liquidity of the swimwear manufacturer will decrease because customers will settle the payment later.
- 2 marks
- 5.7** Factors that the tea beverage store would consider include:
- Lead time: For example, the number of days needed for the tea leaves to be received after the tea beverage store places an order. The longer the lead time, the higher the re-order level.
 - Daily usage or sales of inventory: For example, the amount of tea leaves that the tea beverage store uses to produce tea beverages every day. The greater the daily usage or sales of inventory, the higher the re-order level.
 - Level of safety stock: For example, the minimum level of inventory that the tea beverage store requires to cope with unexpected situations such as a sudden increase in sales. The higher the level of safety stock a firm requires, the higher the re-order level.
- Any two
2 marks each
- 5.8** (a) Inventory turnover
- $= \$2,450,000 \div [(\$880,000 + \$930,000) \div 2]$
- 1 mark
- $= 2.71 \text{ times}$
- 1 mark
- (b) The company may improve its inventory turnover in the following ways:
- Sell more popular products
 - Improve its inventory management so as to reduce the amount of inventory
 - Spend more money on advertisements or organise more sales promotion activities
- Any two
2 marks each
- (Any other reasonable answers)

5.9 (a) Cost (\$)



(1 mark for the graph and 1 mark for the correct labels)

(b) The tuck shop's optimal order quantity of potato chips

$$= \sqrt{\frac{2(20)(200)}{20}}$$

1 mark

$$= 20 \text{ boxes}$$

1 mark

5.10 (a) Re-order level of rice

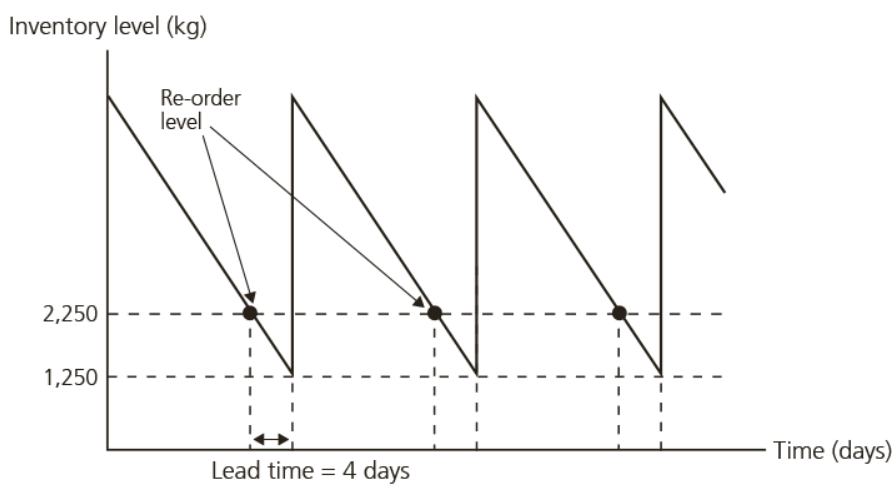
$$= (250 \times 4) + (250 \times 5)$$

1 mark

$$= 2,250 \text{ kg}$$

1 mark

(b)



(1 mark for the graph, 1 mark for the correct labels)

Essay Questions

- 5.11** (a) A cash budget is a budget for both cash receipts and payments. It forecasts a firm's future cash inflows, outflows and cash balance. 2 marks
- This allows the fitness centre to match the timing and the amount of cash outflows with inflows. 2 marks
- As a result, it can:
- plan in advance for short-term investment in the case of a cash surplus. This provides the fitness centre with more time to analyse different investment opportunities and make better decisions; and 2 marks
 - plan in advance for short-term loans in case of a cash deficiency. This gives the fitness centre enough time to search for different financing options and negotiate a loan. 2 marks
- (b) The steps involved in evaluating the performances of the course instructors include:
- Step 1: Set performance standards and communicate with employees
The fitness centre sets the performance standards to be used in the performance appraisals. The standards should be able to direct the efforts of course instructors towards organisational objectives. For example, if the company's objective is to increase revenue, it may set a performance standard for course instructors to achieve a certain number of bookings for training lessons per month. The company should then communicate the performance standards to the course instructors and make sure they understand and agree with the standards.
 - Step 2: Measure actual performance and compare it with performance standards
The fitness centre measures the actual performance of its course instructors according to performance standards. For example, it can measure the number of bookings that each course instructor gets per month by obtaining sales data from the sales record. It can then compare the numbers against the performance standards to check whether course instructors have met the standards.
 - Step 3: Conduct appraisal interviews and provide feedback to employees
The fitness centre conducts appraisal interviews with its course instructors to provide them with feedback based on the appraisal results and to listen to their responses. For example, it may praise the excellent performance of course instructors.
 - Step 4: Make human resources management (HRM) decisions
Based on the performance appraisal results, the fitness centre can make many important HRM decisions. For example, if a course instructor has exceeded the performance standards, the company may promote him or her (manpower planning decision).
- (1 mark for each step and 2 marks for the explanation/example, maximum 10 marks)
- (Any other reasonable examples)

- 5.12** (a) The following explains how a bank may use the 5Cs of credit analysis to assess loan applications:
- **Capital:** The bank may examine the amount of capital the business owner has invested in the business by examining the gearing ratio of the business. This shows the proportion of the business's long-term liabilities relative to long-term capital, including owners' investment.
 - **Capacity:** The bank may examine the ability of the business to repay the debt by examining the amount and timing of cash flows and sales record of the business.
 - **Collateral:** The bank may determine whether or not the business has sufficient assets to secure the loan in the event of default, by examining the value of the asset offered as collateral by the business, such as a property or an automobile.
 - **Conditions:** The bank may consider the overall economic conditions, the industry outlook of the business, as well as the intended purpose of the loan. For example, it may examine how much the business is adversely affected during an economic recession by analysing whether the unemployment rate affects its customers' consumption behaviour.
 - **Character:** The bank may make a judgment regarding the creditworthiness of the business by examining the credit and payment history of the business.
- (1 mark for each factor and 1 mark for the example)
(Any other reasonable examples)

- (b) Advantages of using internal rate of return for capital investment appraisal:
- Time value of money is considered. All cash flows that occur in different years are discounted and thus directly comparable.
 - All relevant cash flows within the investment period are considered, even cash flows beyond the payback period.
 - It is easy to understand and interpret, as it is expressed as a percentage. This helps managers communicate the idea of project profitability to shareholders.

Disadvantages of using internal rate of return for capital investment appraisal:

- Since it is expressed as a percentage, it does not reflect the effect of the project on the firm value. Therefore, when comparing two mutually exclusive projects, the manager cannot assess how much the two projects will increase the firm value.
- Multiple internal rates of return may exist when there are negative cash flows. Mathematically, it requires a set of positive cash flows to generate a single internal rate of return. Therefore, when cash flows include both positive and negative figures, there will be multiple solutions for the internal rate of return.

(2 marks for each point, maximum 8 marks)

- 5.13** (a) Advantages of using channel members for the soft drink manufacturer:
- **Gain access to channel members' expertise and resources:** Channel members usually have good knowledge about their respective markets and possess the expertise and resources required for effective distribution. It may be more cost effective for the soft drink manufacturer to make use of channel members.

- May reach more consumers: By making use of the distribution networks and customer bases of channel members, the soft drink manufacturer may be able to reach more consumers. Channel members are especially useful if the manufacturer's target consumers are geographically dispersed.
- Can focus on core tasks: As channel members will take care of all the distribution functions, the soft drink manufacturer will have less of a workload and can focus on its production tasks.

Disadvantages of using channel members for the soft drink manufacturer:

- Higher distribution costs: The soft drink manufacturer needs to pay channel members for their distribution services. Some channel members may also share the manufacturer's profits.
- Loss of control over the marketing of products: The soft drink manufacturer cannot have complete control over the display, promotion, quality and pricing of its products. For example, channel members may not actively promote the manufacturer's products, or they may store or transport the products in less desirable conditions.
- Cannot reach customers directly: Since the soft drink manufacturer can only reach its customers through channel members, it cannot receive direct feedback from customers and respond immediately. It may find it more difficult to build close relationships with them.

(2 marks for each point, maximum 8 marks)

(b) Factors that the soft drink manufacturer should consider include:

- Liquidity: If the soft drink manufacturer extends too much credit to customers or has too many sales on credit, its money will be tied up in accounts receivable. As a result, its cash position worsens and liquidity decreases. Therefore, if it wants to improve its cash position and liquidity, it should grant less credit and speed up the collection of accounts receivable. For example, the manufacturer may raise credit standards, adjust credit terms to push for and encourage early repayment and implement a stricter collection policy.
- Sales: If the soft drink manufacturer wants to boost sales, it should loosen its accounts receivable policy. This helps firms attract customers from competitors which have a stricter accounts receivable policy. For example, the manufacturer may lower credit standards, adjust credit terms to provide more attractive terms, and implement a more lenient collection policy.
- Bad debts: If the soft drink manufacturer extends too much credit, especially to customers with poor creditworthiness, its risk of bad debts will increase. Therefore, if the manufacturer wants to reduce bad debts, it may raise credit standards, adjust credit terms to push for and encourage early repayment, and implement a stricter collection policy.

(1 mark for each factor and 3 marks for the explanation and example, maximum 10 marks)

5.14 (a) Variables of the EOQ formula include:

- Demand for goods per period: This refers to the quantity of goods that the sporting goods retailer needs in a certain period (normally a year). For example, if the sporting goods retailer sells 1,000 badminton rackets a month, its demand for badminton rackets per year is 12,000 units.
- Ordering costs (fixed cost) per order: They refer to costs related to placing and receiving an order for inventory from suppliers which are normally expressed as a fixed amount per order. For example, they include the costs incurred from the clerical and administrative work involved and transportation costs.
- Carrying costs per unit per period: Carrying costs refer to costs related to holding inventory for a certain time period (normally a year), which are normally measured per unit of inventory. They include storage expenses, insurance costs, the loss due to inventory deterioration and obsolescence, and the opportunity cost of the money tied up in inventory.
- Economic order quantity: This is the order quantity at which the total inventory cost is at the minimum. At this point, the total ordering cost and the total carrying cost are equal. For example, if the economic order quantity of badminton rackets for the sporting goods retailer is 120 units, the retailer should buy 120 badminton rackets per order in order to minimise the total inventory cost.

(1 mark for each variable and 2 marks for the explanation and example, maximum 10 marks)

(Any other reasonable examples)

(b) The sporting goods retailer may acquire competitive advantages by differentiating itself from its competitors in the following ways:

- Product differentiation: This means that a company differentiates itself by its physical product. For example, the sporting goods retailer may become the sole agent for a well-known sports brand so that Hong Kong consumers can only buy that brand's goods from the company.
- Service differentiation: This means that a company differentiates itself by quality services. For example, the sporting goods retailer may protect customers' interests and implement a more lenient return policy than its competitors. It may allow customers to return, exchange and refund products within 30 days if they can provide the official receipts.
- People differentiation: This means that a company differentiates itself by possessing well-trained personnel. For example, the sporting goods retailer may ensure that all of its salespeople are well-trained and can quickly respond to customer enquiries about how to use certain sports equipment.
- Channel differentiation: This means that a company differentiates itself by means of delivering products to its target consumers. For example, the sporting goods retailer may allow customers to pick up the products they ordered online at designated branches.
- Image differentiation: This means that a company differentiates itself by its existing corporate or brand image. For example, the sporting goods retailer may hire a famous athlete as the spokesperson for its sporting goods to enhance its professional image.

(1 mark for the type of differentiation and 2 marks for the explanation/example, maximum 8 marks)

Case Analysis

5.15 (a) (i) Goodday Brewery may boost sales using elements of the credit policy in the following ways:

- Credit standards: Goodday Brewery may lower the credit standards such that credit will be granted to customers more easily. As a result, more customers will be attracted into buying from Goodday Brewery.
- Credit terms: Goodday Brewery may attract more customers into buying from the firm by providing more attractive terms. For example, it may provide greater payment flexibility to customers by extending the cash discount period. Also, it may increase the cash discount it offers to customers, such that more customers (cash discount takers) are attracted by the higher cash discount rate.
- Collection policy: Goodday Brewery may implement a more lenient collection policy, such as reaching out to debtors less frequently and taking more amicable debt collection actions. As such, more customers will be attracted by the lenient collection policy.

Any two

2 marks each

(Any other reasonable examples)

(ii) Drawbacks of extending the credit period for Goodday Brewery:

- Its cash position will worsen and its liquidity will decrease. This is because customers will settle their payment later.
- Its bad debts will increase. This is because customers will owe the firm money for a longer period of time, and the chance that some of the customers will have financial difficulties and not be able to repay their debt will therefore become greater.

2 marks

2 marks

(b) For product strategy, Goodday Brewery may expand its product line by adding product items for consumers that are not currently served by the company, such as alcohol-free beers to serve more customers and generate more sales revenue.

2 marks

For promotion strategy, Goodday Brewery may use sales promotion tools such as discount coupons to boost sales.

2 marks

(Any other reasonable examples)

(c) Role playing

1 mark

Advantages of role playing:

- Useful for developing interpersonal and communication skills: By interacting with each other in role plays, Goodday Brewery's staff members can learn interpersonal and communication skills more effectively.
- Prepare for real-life situations: Role playing can help Goodday Brewery's staff members prepare for situations that they may encounter during their jobs. This can help them handle similar real-life situations more effectively.

Any

3 marks

(Any other reasonable answers)

5.16 (a) Objectives of inventory management include:

- maintaining an adequate level of inventory 1 mark
- minimising the costs associated with inventory 1 mark

(b) (i) Re-order level is the level of inventory at which a firm should place an order to replenish inventory. 2 marks

(ii) Factors that would affect the re-order level of soybeans:

- Lead time: The longer the lead time, the higher the re-order level of soybeans. 2 marks
- Daily sales or usage of inventory: The greater the daily usage or sales of soybeans, the higher the re-order level. 2 marks

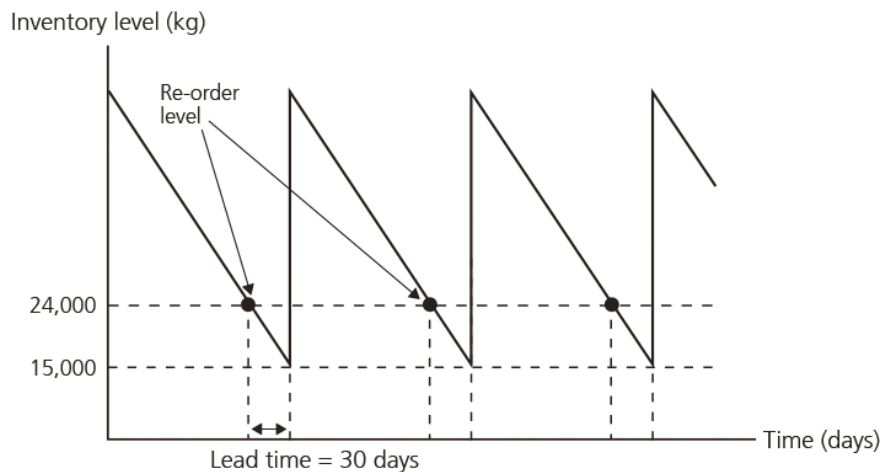
(c) Re-order level of soybeans for Tai Wo Soy Factory

$$= (300 \times 30) + (300 \times 50)$$

1 mark

$$= 24,000 \text{ kg}$$

1 mark



(1 mark for the graph, 1 mark for the correct labels)

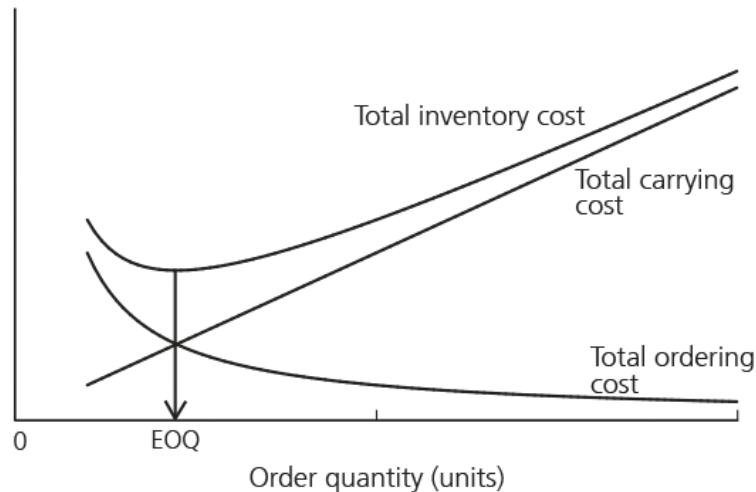
(d) The type of risk management strategy that Martin suggested using is risk reduction (loss prevention). It involves measures that can help prevent losses from happening, i.e., reduce loss frequency. 2 marks

(e) Limitations of using the payback period to assess capital investment projects for Tai Wo Soy Factory:

- It ignores the time value of money.
- It ignores cash flows after the payback period. Any
- The decision criterion for a targeted payback period is rather subjective and arbitrary. 2 marks

(Any other reasonable answers)

5.17 (a) Cost (\$)



(1 mark for the graph and 1 mark for the correct labels)

The total inventory cost is at a minimum when the total ordering cost and total carrying cost curves intersect each other.

1 mark

Therefore, the economic order quantity is arrived at when the total ordering cost is equal to total carrying cost.

1 mark

(b) Economic order quantity of portable hard drives for COMP Limited

$$= \sqrt{\frac{2(18,000)(500)}{80}}$$

1 mark

$$= 475 \text{ units}$$

1 mark

(c) COMP Limited may lower the risk of bad debts in the following ways:

- Credit standards: COMP Limited may raise its credit standards.
- Credit terms: COMP Limited may shorten the credit period and increase the cash discount.
- Collection policy: COMP Limited may implement a stricter collection policy.

Any two

2 marks each

(Any other reasonable examples)

(d) Advantages of using a monthly salary to reward salespeople for COMP Limited:

- Less competition among employees: As employees are not compensated for performance, competition among employees is reduced. This helps develop a harmonious working environment
- Simple and cost saving: Time-based pay is simple and easy to calculate. Companies can avoid the difficulties and administrative costs involved in evaluating employees' performance.
- Facilitate planning and budgeting: Rewarding employees with time-based pay makes it easier for companies to plan and budget for labour costs.

Any

2 marks

(e) Reasons to support Judy's suggestion of using commissions to reward salespeople:

- Simple: It is easy to understand and apply.
- Strong motivational power: It provides the greatest incentive for employees to work hard as they can easily see the link between effort and reward.
- Help increase sales revenue: As employees will work hard to earn more commissions, this helps increase the company's sales revenue.

Any two
2 marks each